Random Variables

Contents

1 Random Variables

When an experiment is performed, we are frequently interested mainly in some function of the outcome as opposed to the actual outcome itself. For instance, in tossing dice, we are often interested in the sum of the two dice and are not really concerned about the separate values of each die. That is, we may be interested in knowing that the sum is 7 and may not be concerned over whether the actual outcome was (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), or (6, 1). Also, in flipping a coin, we may be interested in the total number of heads that occur and not care at all about the actual head–tail sequence that results. These quantities of interest, or, more formally, these real-valued functions defined on the sample space, are known as *random variables*.

Because the value of a random variable is determined by the outcome of the experiment, we may assign probabilities to the possible values of the random variable.

Example 1a Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2, and 3 with respective probabilities

$$P\{Y = 0\} = P\{(T, T, T)\} = \frac{1}{8}$$

$$P\{Y = 1\} = P\{(T, T, H), (T, H, T), (H, T, T)\} = \frac{3}{8}$$

$$P\{Y = 2\} = P\{(T, H, H), (H, T, H), (H, H, T)\} = \frac{3}{8}$$

$$P\{Y = 3\} = P\{(H, H, H)\} = \frac{1}{8}$$

Since Y must take on one of the values 0 through 3, we must have

$$1 = P\left(\bigcup_{i=0}^{3}\{Y = i\}\right) = \sum_{i=0}^{3} P\{Y = i\}$$

which, of course, is in accord with the preceding probabilities. ■

Example 1b

A life insurance agent has 2 elderly clients, each of whom has a life insurance policy that pays \$100,000 upon death. Let Y be the event that the younger one dies in the following year, and let O be the event that the older one dies in the following year. Assume that Y and O are independent, with respective probabilities $P(Y) = .05$ and $P(O) = .10$. If X denotes the total amount of money (in units of \$100,000) that will be paid out this year to any of these clients' beneficiaries, then X is a random variable that takes on one of the possible values $0, 1, 2$ with respective probabilities

$$P\{X = 0\} = P(Y^cO^c) = P(Y^c)P(O^c) = (.95)(.9) = .855$$
$$P\{X = 1\} = P(YO^c) + P(Y^cO) = (.05)(.9) + (.95)(.1) = .140$$
$$P\{X = 2\} = P(YO) = (.05)(.1) = .005$$ ■

Example 1c

Four balls are to be randomly selected, without replacement, from an urn that contains 20 balls numbered 1 through 20. If X is the largest numbered ball selected, then X is a random variable that takes on one of the values $4, 5, \ldots, 20$. Because each of the $\binom{20}{4}$ possible selections of 4 of the 20 balls is equally likely, the probability that X takes on each of its possible values is

$$P\{X = i\} = \frac{\binom{i-1}{3}}{\binom{20}{4}}, \quad i = 4, \ldots, 20$$

This is so because the number of selections that result in $X = i$ is the number of selections that result in ball numbered i and three of the balls numbered 1 through $i - 1$ being selected. As there are $\binom{1}{1}\binom{i-1}{3}$ such selections, the preceding equation follows.

Suppose now that we want to determine $P\{X > 10\}$. One way, of course, is to just use the preceding to obtain

$$P\{X > 10\} = \sum_{i=11}^{20} P\{X = i\} = \sum_{i=11}^{20} \frac{\binom{i-1}{3}}{\binom{20}{4}}$$

However, a more direct approach for determining $P(X > 10)$ would be to use

$$P\{X > 10\} = 1 - P\{X \le 10\} = 1 - \frac{\binom{10}{4}}{\binom{20}{4}}$$

where the preceding results because X will be less than or equal to 10 when the 4 balls chosen are among balls numbered 1 through 10. ■

Example 1d

Independent trials consisting of the flipping of a coin having probability p of coming up heads are continually performed until either a head occurs or a total of n flips is made. If we let X denote the number of times the coin is flipped, then X is a random variable taking on one of the values $1, 2, 3, \ldots, n$ with respective probabilities

$$P\{X = 1\} = P\{H\} = p$$

$$P\{X = 2\} = P\{(T, H)\} = (1 - p)p$$

$$P\{X = 3\} = P\{(T, T, H)\} = (1 - p)^2 p$$

$$\vdots$$

$$P\{X = n - 1\} = P\{(\underbrace{T, T, \ldots, T}_{n-2}, H)\} = (1 - p)^{n-2} p$$

$$P\{X = n\} = P\{(\underbrace{T, T, \ldots, T}_{n-1}, T), (\underbrace{T, T, \ldots, T}_{n-1}, H)\} = (1 - p)^{n-1}$$

As a check, note that

$$\begin{aligned} P\left(\bigcup_{i=1}^{n} \{X = i\}\right) &= \sum_{i=1}^{n} P\{X = i\} \\ &= \sum_{i=1}^{n-1} p(1 - p)^{i-1} + (1 - p)^{n-1} \\ &= p\left[\frac{1 - (1 - p)^{n-1}}{1 - (1 - p)}\right] + (1 - p)^{n-1} \\ &= 1 - (1 - p)^{n-1} + (1 - p)^{n-1} \\ &= 1 \end{aligned}$$

■

Example 1e

Suppose that there are N distinct types of coupons and that each time one obtains a coupon, it is, independently of previous selections, equally likely to be any one of the N types. One random variable of interest is T, the number of coupons that need to be collected until one obtains a complete set of at least one of each type. Rather than derive $P\{T = n\}$ directly, let us start by considering the probability that T is greater than n. To do so, fix n and define the events $A_1, A_2, \ldots, A_N$ as follows: A_j is the event that no type j coupon is contained among the first n coupons collected, $j = 1, \ldots, N$. Hence,

$$\begin{aligned} P\{T > n\} &= P\left(\bigcup_{j=1}^{N} A_j\right) \\ &= \sum_j P(A_j) - \sum\sum_{j_1 < j_2} P(A_{j_1} A_{j_2}) + \cdots \\ &\quad + (-1)^{k+1} \sum\sum\sum_{j_1 < j_2 < \cdots < j_k} P(A_{j_1} A_{j_2} \cdots A_{j_k}) \cdots \\ &\quad + (-1)^{N+1} P(A_1 A_2 \cdots A_N) \end{aligned}$$

Now, A_j will occur if each of the n coupons collected is not of type j. Since each of the coupons will not be of type j with probability $(N - 1)/N$, we have, by the assumed independence of the types of successive coupons,

$$P(A_j) = \left(\frac{N-1}{N}\right)^n$$

Also, the event $A_{j_1}A_{j_2}$ will occur if none of the first n coupons collected is of either type j_1 or type j_2. Thus, again using independence, we see that

$$P(A_{j_1}A_{j_2}) = \left(\frac{N-2}{N}\right)^n$$

The same reasoning gives

$$P(A_{j_1}A_{j_2}\cdots A_{j_k}) = \left(\frac{N-k}{N}\right)^n$$

and we see that for $n > 0$,

$$\begin{aligned}
P\{T > n\} &= N\left(\frac{N-1}{N}\right)^n - \binom{N}{2}\left(\frac{N-2}{N}\right)^n + \binom{N}{3}\left(\frac{N-3}{N}\right)^n - \cdots \\
&\quad + (-1)^N \binom{N}{N-1}\left(\frac{1}{N}\right)^n \\
&= \sum_{i=1}^{N-1}\binom{N}{i}\left(\frac{N-i}{N}\right)^n (-1)^{i+1} \qquad (1.1)
\end{aligned}$$

The probability that T equals n can now be obtained from the preceding formula by the use of

$$P\{T > n - 1\} = P\{T = n\} + P\{T > n\}$$

or, equivalently,

$$P\{T = n\} = P\{T > n - 1\} - P\{T > n\}$$

Another random variable of interest is the number of distinct types of coupons that are contained in the first n selections—call this random variable D_n. To compute $P\{D_n = k\}$, let us start by fixing attention on a particular set of k distinct types, and let us then determine the probability that this set constitutes the set of distinct types obtained in the first n selections. Now, in order for this to be the situation, it is necessary and sufficient that of the first n coupons obtained,

A: each is one of these k types

B: each of these k types is represented

Now, each coupon selected will be one of the k types with probability k/N, so the probability that A will be valid is $(k/N)^n$. Also, given that a coupon is of one of the k types under consideration, it is easy to see that it is equally likely to be of any one of these k types. Hence, the conditional probability of B given that A occurs is the same as the probability that a set of n coupons, each equally likely to be any of k possible types, contains a complete set of all k types. But this is just the probability that the number needed to amass a complete set, when choosing among k types, is less than or equal to n and is thus obtainable from Equation (1.1) with k replacing N. Thus, we have

$$P(A) = \left(\frac{k}{N}\right)^n$$

$$P(B|A) = 1 - \sum_{i=1}^{k-1} \binom{k}{i} \left(\frac{k-i}{k}\right)^n (-1)^{i+1}$$

Finally, as there are $\binom{N}{k}$ possible choices for the set of k types, we arrive at

$$\begin{aligned} P\{D_n = k\} &= \binom{N}{k} P(AB) \\ &= \binom{N}{k} \left(\frac{k}{N}\right)^n \left[1 - \sum_{i=1}^{k-1} \binom{k}{i} \left(\frac{k-i}{k}\right)^n (-1)^{i+1}\right] \end{aligned}$$

Remark Since one must collect at least N coupons to obtain a complete set, it follows that $P\{T > n\} = 1$ if $n < N$. Therefore, from Equation (1.1), we obtain the interesting combinatorial identity that for integers $1 \leq n < N$,

$$\sum_{i=1}^{N-1} \binom{N}{i} \left(\frac{N-i}{N}\right)^n (-1)^{i+1} = 1$$

which can be written as

$$\sum_{i=0}^{N-1} \binom{N}{i} \left(\frac{N-i}{N}\right)^n (-1)^{i+1} = 0$$

or, upon multiplying by $(-1)^N N^n$ and letting $j = N - i$,

$$\sum_{j=1}^{N} \binom{N}{j} j^n (-1)^{j-1} = 0 \qquad 1 \leq n < N$$ ■

For a random variable X, the function F defined by

$$F(x) = P\{X \leq x\} \qquad -\infty < x < \infty$$

is called the *cumulative distribution function* or, more simply, the *distribution function* of X. Thus, the distribution function specifies, for all real values x, the probability that the random variable is less than or equal to x.

Now, suppose that $a \leq b$. Then, because the event $\{X \leq a\}$ is contained in the event $\{X \leq b\}$, it follows that $F(a)$, the probability of the former, is less than or equal to $F(b)$, the probability of the latter. In other words, $F(x)$ is a nondecreasing function of x. Other general properties of the distribution function are given in Section 10.

2 Discrete Random Variables

A random variable that can take on at most a countable number of possible values is said to be discrete. For a discrete random variable X, we define the *probability mass function* $p(a)$ of X by

$$p(a) = P\{X = a\}$$

The probability mass function $p(a)$ is positive for at most a countable number of values of a. That is, if X must assume one of the values $x_1, x_2, \ldots$, then

$$\begin{aligned} p(x_i) &\geq 0 \quad \text{for } i = 1, 2, \ldots \\ p(x) &= 0 \quad \text{for all other values of } x \end{aligned}$$

Since X must take on one of the values x_i, we have

$$\sum_{i=1}^{\infty} p(x_i) = 1$$

It is often instructive to present the probability mass function in a graphical format by plotting $p(x_i)$ on the y-axis against x_i on the x-axis. For instance, if the probability mass function of X is

$$p(0) = \frac{1}{4} \qquad p(1) = \frac{1}{2} \qquad p(2) = \frac{1}{4}$$

we can represent this function graphically as shown in Figure 1. Similarly, a graph of the probability mass function of the random variable representing the sum when two dice are rolled looks like Figure 2.

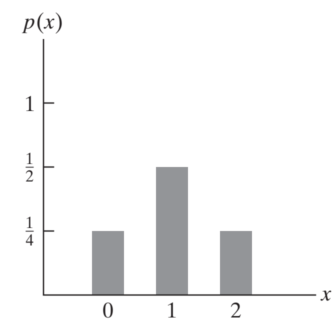


Figure 1

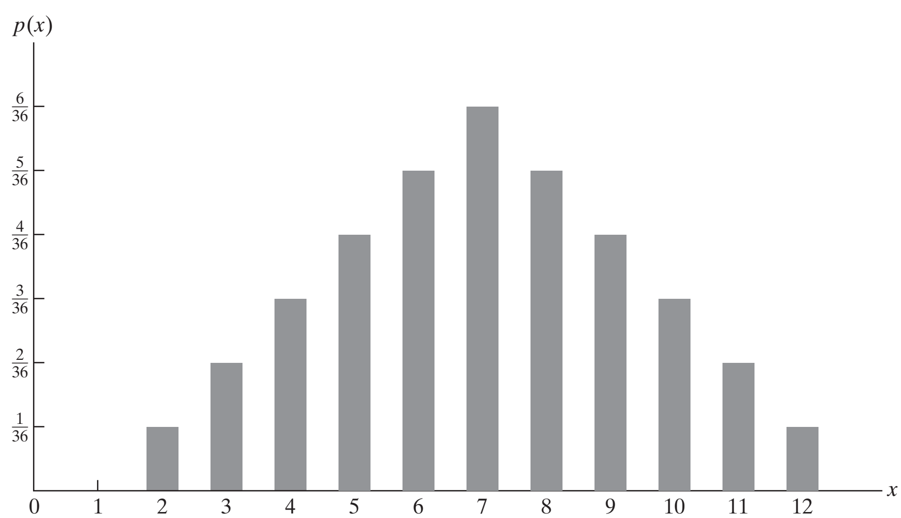


Figure 2

Example 2a The probability mass function of a random variable X is given by $p(i) = c\lambda^i/i!$, $i = 0, 1, 2, \ldots$, where λ is some positive value. Find (a) $P\{X = 0\}$ and (b) $P\{X > 2\}$.

Solution Since $\sum_{i=0}^{\infty} p(i) = 1$, we have

$$c\sum_{i=0}^{\infty}\frac{\lambda^i}{i!} = 1$$

which, because $e^x = \sum_{i=0}^{\infty} x^i/i!$, implies that

$$ce^{\lambda} = 1 \quad \text{or} \quad c = e^{-\lambda}$$

Hence,

(a) $P\{X = 0\} = e^{-\lambda}\lambda^0/0! = e^{-\lambda}$

(b) $$\begin{aligned} P\{X > 2\} &= 1 - P\{X \le 2\} = 1 - P\{X = 0\} - P\{X = 1\} \\ &\quad - P\{X = 2\} \\ &= 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2} \end{aligned}$$ ■

The cumulative distribution function F can be expressed in terms of $p(a)$ by

$$F(a) = \sum_{\text{all } x \le a} p(x)$$

If X is a discrete random variable whose possible values are $x_1, x_2, x_3, \ldots$, where $x_1 < x_2 < x_3 < \cdots$, then the distribution function F of X is a step function. That is, the value of F is constant in the intervals (x_{i-1}, x_i) and then takes a step (or jump) of size $p(x_i)$ at x_i. For instance, if X has a probability mass function given by

$$p(1) = \frac{1}{4} \qquad p(2) = \frac{1}{2} \qquad p(3) = \frac{1}{8} \qquad p(4) = \frac{1}{8}$$

then its cumulative distribution function is

$$F(a) = \begin{cases} 0 & a < 1 \\ \frac{1}{4} & 1 \le a < 2 \\ \frac{3}{4} & 2 \le a < 3 \\ \frac{7}{8} & 3 \le a < 4 \\ 1 & 4 \le a \end{cases}$$

This function is depicted graphically in Figure 3.

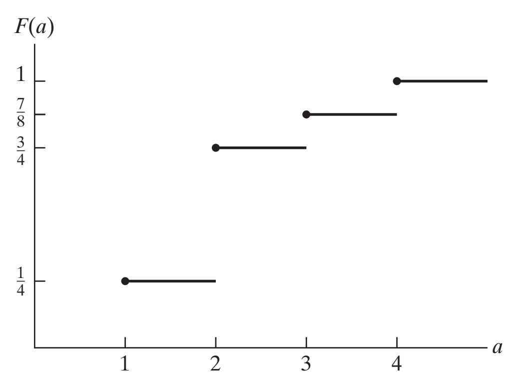


Figure 3

Note that the size of the step at any of the values 1, 2, 3, and 4 is equal to the probability that X assumes that particular value.

3 Expected Value

One of the most important concepts in probability theory is that of the expectation of a random variable. If X is a discrete random variable having a probability mass function $p(x)$, then the *expectation*, or the *expected value*, of X, denoted by $E[X]$, is defined by

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

In words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it. For instance, on the one hand, if the probability mass function of X is given by

$$p(0) = \frac{1}{2} = p(1)$$

then

$$E[X] = 0\left(\frac{1}{2}\right) + 1\left(\frac{1}{2}\right) = \frac{1}{2}$$

is just the ordinary average of the two possible values, 0 and 1, that X can assume. On the other hand, if

$$p(0) = \frac{1}{3} \qquad p(1) = \frac{2}{3}$$

then

$$E[X] = 0\left(\frac{1}{3}\right) + 1\left(\frac{2}{3}\right) = \frac{2}{3}$$

is a weighted average of the two possible values 0 and 1, where the value 1 is given twice as much weight as the value 0, since $p(1) = 2p(0)$.

Another motivation of the definition of expectation is provided by the frequency interpretation of probabilities. This interpretation (partially justified by the strong law of large numbers) assumes that if an infinite sequence of independent replications of an experiment is performed, then, for any event E, the proportion of time that E occurs will be $P(E)$. Now, consider a random variable X that must take on one of the values $x_1, x_2, \ldots x_n$ with respective probabilities $p(x_1), p(x_2), \ldots, p(x_n)$, and think of X as representing our winnings in a single game of chance. That is, with probability $p(x_i)$, we shall win x_i units $i = 1, 2, \ldots, n$. By the frequency interpretation, if we play this game continually, then the proportion of time that we win x_i will be $p(x_i)$. Since this is true for all $i, i = 1, 2, \ldots, n$, it follows that our average winnings per game will be

$$\sum_{i=1}^{n} x_i p(x_i) = E[X]$$

Example 3a Find $E[X]$, where X is the outcome when we roll a fair die.

Solution Since $p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6}$, we obtain

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$ ■

Example 3b We say that I is an indicator variable for the event A if

$$I = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{if } A^c \text{ occurs} \end{cases}$$

Find $E[I]$.

Solution Since $p(1) = P(A), p(0) = 1 - P(A)$, we have

$$E[I] = P(A)$$

That is, the expected value of the indicator variable for the event A is equal to the probability that A occurs. ■

Example 3c A contestant on a quiz show is presented with two questions, questions 1 and 2, which he is to attempt to answer in some order he chooses. If he decides to try question i first, then he will be allowed to go on to question $j, j \neq i$, only if his answer to question i is correct. If his initial answer is incorrect, he is not allowed to answer the other question. The contestant is to receive V_i dollars if he answers question i correctly, $i = 1, 2$. For instance, he will receive $V_1 + V_2$ dollars if he answers both questions correctly. If the probability that he knows the answer to question i is $P_i, i = 1, 2$, which question should he attempt to answer first so as to maximize his expected winnings? Assume that the events $E_i, i = 1, 2$, that he knows the answer to question i are independent events.

Solution On the one hand, if he attempts to answer question 1 first, then he will win

$$\begin{array}{ll} 0 & \text{with probability } 1 - P_1 \\ V_1 & \text{with probability } P_1(1 - P_2) \\ V_1 + V_2 & \text{with probability } P_1 P_2 \end{array}$$

Hence, his expected winnings in this case will be

$$V_1 P_1(1 - P_2) + (V_1 + V_2)P_1 P_2$$

On the other hand, if he attempts to answer question 2 first, his expected winnings will be

$$V_2 P_2(1 - P_1) + (V_1 + V_2)P_1 P_2$$

Therefore, it is better to try question 1 first if

$$V_1 P_1(1 - P_2) \geq V_2 P_2(1 - P_1)$$

or, equivalently, if

$$\frac{V_1 P_1}{1 - P_1} \geq \frac{V_2 P_2}{1 - P_2}$$

For example, if he is 60 percent certain of answering question 1, worth \$200, correctly and he is 80 percent certain of answering question 2, worth \$100, correctly, then he should attempt to answer question 2 first because

$$400 = \frac{(100)(.8)}{.2} > \frac{(200)(.6)}{.4} = 300$$ ■

Example 3d A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E[X]$.

Solution Since the randomly chosen student is equally likely to be any of the 120 students, it follows that

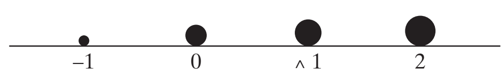
$$P\{X = 36\} = \frac{36}{120} \quad P\{X = 40\} = \frac{40}{120} \quad P\{X = 44\} = \frac{44}{120}$$

Hence,

$$E[X] = 36\left(\frac{3}{10}\right) + 40\left(\frac{1}{3}\right) + 44\left(\frac{11}{30}\right) = \frac{1208}{30} = 40.2667$$

However, the average number of students on a bus is $120/3 = 40$, showing that the expected number of students on the bus of a randomly chosen student is larger than the average number of students on a bus. This is a general phenomenon, and it occurs because the more students there are on a bus, the more likely it is that a randomly chosen student would have been on that bus. As a result, buses with many students are given more weight than those with fewer students. (See Self-Test Problem 4) ■

Remark The probability concept of expectation is analogous to the physical concept of the *center of gravity* of a distribution of mass. Consider a discrete random variable X having probability mass function $p(x_i), i \ge 1$. If we now imagine a weightless rod in which weights with mass $p(x_i), i \ge 1$, are located at the points $x_i, i \ge 1$ (see Figure 4), then the point at which the rod would be in balance is known as the center of gravity. For those readers acquainted with elementary statics, it is now a simple matter to show that this point is at $E[X]$.† ■



$p(-1) = .10, \quad p(0) = .25, \quad p(1) = .30, \quad p(2) = .35$

$\wedge$ = center of gravity = .9

Figure 4

4 Expectation of a Function of a Random Variable

Suppose that we are given a discrete random variable along with its probability mass function and that we want to compute the expected value of some function of X, say, $g(X)$. How can we accomplish this? One way is as follows: Since $g(X)$ is itself a discrete random variable, it has a probability mass function, which can be determined from the probability mass function of X. Once we have determined the probability mass function of $g(X)$, we can compute $E[g(X)]$ by using the definition of expected value.

†To prove this, we must show that the sum of the torques tending to turn the point around $E[X]$ is equal to 0. That is, we must show that $0 = \sum_i (x_i - E[X])p(x_i)$, which is immediate.

Example 4a Let X denote a random variable that takes on any of the values -1, 0, and 1 with respective probabilities

$$P\{X = -1\} = .2 \quad P\{X = 0\} = .5 \quad P\{X = 1\} = .3$$

Compute $E[X^2]$.

Solution Let $Y = X^2$. Then the probability mass function of Y is given by

$$\begin{aligned} P\{Y = 1\} &= P\{X = -1\} + P\{X = 1\} = .5 \\ P\{Y = 0\} &= P\{X = 0\} = .5 \end{aligned}$$

Hence,

$$E[X^2] = E[Y] = 1(.5) + 0(.5) = .5$$

Note that

$$.5 = E[X^2] \neq (E[X])^2 = .01$$

■

Although the preceding procedure will always enable us to compute the expected value of any function of X from a knowledge of the probability mass function of X, there is another way of thinking about $E[g(X)]$: Since $g(X)$ will equal $g(x)$ whenever X is equal to x, it seems reasonable that $E[g(X)]$ should just be a weighted average of the values $g(x)$, with $g(x)$ being weighted by the probability that X is equal to x. That is, the following result is quite intuitive.

Proposition 4.1 If X is a discrete random variable that takes on one of the values $x_i, i \geq 1$, with respective probabilities $p(x_i)$, then, for any real-valued function g,

$$E[g(X)] = \sum_i g(x_i)p(x_i)$$

Before proving this proposition, let us check that it is in accord with the results of Example 4a. Applying it to that example yields

$$\begin{aligned} E\{X^2\} &= (-1)^2(.2) + 0^2(.5) + 1^2(.3) \\ &= 1(.2 + .3) + 0(.5) \\ &= .5 \end{aligned}$$

which is in agreement with the result given in Example 4a.

Proof of Proposition 4.1 The proof of Proposition 4.1 proceeds, as in the preceding verification, by grouping together all the terms in $\sum_i g(x_i)p(x_i)$ having the same value of $g(x_i)$. Specifically, suppose that $y_j, j \geq 1$, represent the different values of $g(x_i), i \geq 1$. Then, grouping all the $g(x_i)$ having the same value gives

$$\begin{aligned} \sum_i g(x_i)p(x_i) &= \sum_j \sum_{i:g(x_i)=y_j} g(x_i)p(x_i) \\ &= \sum_j \sum_{i:g(x_i)=y_j} y_j p(x_i) \\ &= \sum_j y_j \sum_{i:g(x_i)=y_j} p(x_i) \\ &= \sum_j y_j P\{g(X) = y_j\} \\ &= E[g(X)] \end{aligned}$$

□

Example 4b A product that is sold seasonally yields a net profit of b dollars for each unit sold and a net loss of ℓ dollars for each unit left unsold when the season ends. The number of units of the product that are ordered at a specific department store during any season is a random variable having probability mass function $p(i), i \geq 0$. If the store must stock this product in advance, determine the number of units the store should stock so as to maximize its expected profit.

Solution Let X denote the number of units ordered. If s units are stocked, then the profit—call it $P(s)$—can be expressed as

$$\begin{aligned} P(s) &= bX - (s - X)\ell \quad \text{if } X \leq s \\ &= sb \qquad\qquad\qquad\quad \text{if } X > s \end{aligned}$$

Hence, the expected profit equals

$$\begin{aligned} E[P(s)] &= \sum_{i=0}^{s}[bi - (s - i)\ell]p(i) + \sum_{i=s+1}^{\infty} sbp(i) \\ &= (b + \ell)\sum_{i=0}^{s} ip(i) - s\ell\sum_{i=0}^{s} p(i) + sb\left[1 - \sum_{i=0}^{s} p(i)\right] \\ &= (b + \ell)\sum_{i=0}^{s} ip(i) - (b + \ell)s\sum_{i=0}^{s} p(i) + sb \\ &= sb + (b + \ell)\sum_{i=0}^{s}(i - s)p(i) \end{aligned}$$

To determine the optimum value of s, let us investigate what happens to the profit when we increase s by 1 unit. By substitution, we see that the expected profit in this case is given by

$$\begin{aligned} E[P(s + 1)] &= b(s + 1) + (b + \ell)\sum_{i=0}^{s+1}(i - s - 1)p(i) \\ &= b(s + 1) + (b + \ell)\sum_{i=0}^{s}(i - s - 1)p(i) \end{aligned}$$

Therefore,

$$E[P(s + 1)] - E[P(s)] = b - (b + \ell)\sum_{i=0}^{s} p(i)$$

Thus, stocking $s + 1$ units will be better than stocking s units whenever

$$\sum_{i=0}^{s} p(i) < \frac{b}{b + \ell} \tag{4.1}$$

Because the left-hand side of Equation (4.1) is increasing in s while the right-hand side is constant, the inequality will be satisfied for all values of $s \leq s^*$, where s^* is the largest value of s satisfying Equation (4.1). Since

$$E[P(0)] < \cdots < E[P(s^*)] < E[P(s^* + 1)] > E[P(s^* + 2)] > \cdots$$

it follows that stocking $s^* + 1$ items will lead to a maximum expected profit. ■

Example 4c **Utility**

Suppose that you must choose one of two possible actions, each of which can result in any of n consequences, denoted as $C_1, \ldots, C_n$. Suppose that if the first action is chosen, then consequence C_i will result with probability $p_i, i = 1, \ldots, n$, whereas if the second action is chosen, then consequence C_i will result with probability q_i, $i = 1, \ldots, n$, where $\sum_{i=1}^{n} p_i = \sum_{i=1}^{n} q_i = 1$. The following approach can be used to determine which action to choose: Start by assigning numerical values to the different consequences. First, identify the least and the most desirable consequences—call them c and C, respectively; give consequence c the value 0 and give C the value 1. Now consider any of the other $n - 2$ consequences, say, C_i. To value this consequence, imagine that you are given the choice between either receiving C_i or taking part in a random experiment that either earns you consequence C with probability u or consequence c with probability $1 - u$. Clearly, your choice will depend on the value of u. On the one hand, if $u = 1$, then the experiment is certain to result in consequence C, and since C is the most desirable consequence, you will prefer participating in the experiment to receiving C_i. On the other hand, if $u = 0$, then the experiment will result in the least desirable consequence—namely, c—so in this case you will prefer the consequence C_i to participating in the experiment. Now, as u decreases from 1 to 0, it seems reasonable that your choice will at some point switch from participating in the experiment to the certain return of C_i, and at that critical switch point you will be indifferent between the two alternatives. Take that indifference probability u as the value of the consequence C_i. In other words, the value of C_i is that probability u such that you are indifferent between either receiving the consequence C_i or taking part in an experiment that returns consequence C with probability u or consequence c with probability $1 - u$. We call this indifference probability the *utility* of the consequence C_i, and we designate it as $u(C_i)$.

To determine which action is superior, we need to evaluate each one. Consider the first action, which results in consequence C_i with probability $p_i, i = 1, \ldots, n$. We can think of the result of this action as being determined by a two-stage experiment. In the first stage, one of the values $1, \ldots, n$ is chosen according to the probabilities $p_1, \ldots, p_n$; if value i is chosen, you receive consequence C_i. However, since C_i is equivalent to obtaining consequence C with probability $u(C_i)$ or consequence c with probability $1 - u(C_i)$, it follows that the result of the two-stage experiment is equivalent to an experiment in which either consequence C or consequence c is obtained, with C being obtained with probability

$$\sum_{i=1}^{n} p_i u(C_i)$$

Similarly, the result of choosing the second action is equivalent to taking part in an experiment in which either consequence C or consequence c is obtained, with C being obtained with probability

$$\sum_{i=1}^{n} q_i u(C_i)$$

Since C is preferable to c, it follows that the first action is preferable to the second action if

$$\sum_{i=1}^{n} p_i u(C_i) > \sum_{i=1}^{n} q_i u(C_i)$$

In other words, the worth of an action can be measured by the expected value of the utility of its consequence, and the action with the largest expected utility is the most preferable. ■

A simple logical consequence of Proposition 4.1 is Corollary 4.1.

Corollary 4.1 If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

Proof

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x) \\ &= aE[X] + b \end{aligned}$$ □

The expected value of a random variable X, $E[X]$, is also referred to as the *mean* or the *first moment of X*. The quantity $E[X^n]$, $n \geq 1$, is called the *nth moment of X*. By Proposition 4.1, we note that

$$E[X^n] = \sum_{x:p(x)>0} x^n p(x)$$

5 Variance

Given a random variable X along with its distribution function F, it would be extremely useful if we were able to summarize the essential properties of F by certain suitably defined measures. One such measure would be $E[X]$, the expected value of X. However, although $E[X]$ yields the weighted average of the possible values of X, it does not tell us anything about the variation, or spread, of these values. For instance, although random variables W, Y, and Z having probability mass functions determined by

$$W = 0 \quad \text{with probability } 1$$

$$Y = \begin{cases} -1 & \text{with probability } \frac{1}{2} \\ +1 & \text{with probability } \frac{1}{2} \end{cases}$$

$$Z = \begin{cases} -100 & \text{with probability } \frac{1}{2} \\ +100 & \text{with probability } \frac{1}{2} \end{cases}$$

all have the same expectation—namely, 0—there is a much greater spread in the possible values of Y than in those of W (which is a constant) and in the possible values of Z than in those of Y.

Because we expect X to take on values around its mean $E[X]$, it would appear that a reasonable way of measuring the possible variation of X would be to look at how far apart X would be from its mean, on the average. One possible way to measure this variation would be to consider the quantity $E[|X - \mu|]$, where $\mu = E[X]$. However, it turns out to be mathematically inconvenient to deal with this quantity, so a more tractable quantity is usually considered—namely, the expectation of the square of the difference between X and its mean. We thus have the following definition.

Definition
If X is a random variable with mean μ, then the variance of X, denoted by $\text{Var}(X)$, is defined by

$$\text{Var}(X) = E[(X - \mu)^2]$$

An alternative formula for $\text{Var}(X)$ is derived as follows:

$$\begin{aligned}
\text{Var}(X) &= E[(X - \mu)^2] \\
&= \sum_x (x - \mu)^2 p(x) \\
&= \sum_x (x^2 - 2\mu x + \mu^2) p(x) \\
&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2
\end{aligned}$$

That is,

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2}$$

In words, the variance of X is equal to the expected value of X^2 minus the square of its expected value. In practice, this formula frequently offers the easiest way to compute $\text{Var}(X)$.

Example 5a

Calculate $\text{Var}(X)$ if X represents the outcome when a fair die is rolled.

Solution It was shown in Example 3a that $E[X] = \frac{7}{2}$. Also,

$$\begin{aligned}
E[X^2] &= 1^2\left(\frac{1}{6}\right) + 2^2\left(\frac{1}{6}\right) + 3^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 5^2\left(\frac{1}{6}\right) + 6^2\left(\frac{1}{6}\right) \\
&= \left(\frac{1}{6}\right)(91)
\end{aligned}$$

Hence,

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

■

A useful identity is that for any constants a and b,

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

To prove this equality, let $\mu = E[X]$ and note from Corollary 4.1 that $E[aX + b] = a\mu + b$. Therefore,

$$\begin{aligned}
\text{Var}(aX + b) &= E[(aX + b - a\mu - b)^2] \\
&= E[a^2(X - \mu)^2] \\
&= a^2 E[(X - \mu)^2] \\
&= a^2\text{Var}(X)
\end{aligned}$$

Remarks (a) Analogous to the means being the center of gravity of a distribution of mass, the variance represents, in the terminology of mechanics, the moment of inertia.

(b) The square root of the Var(X) is called the *standard deviation* of X, and we denote it by SD(X). That is,

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

Discrete random variables are often classified according to their probability mass functions. In the next few sections, we consider some of the more common types.

6 The Bernoulli and Binomial Random Variables

Suppose that a trial, or an experiment, whose outcome can be classified as either a *success* or a *failure* is performed. If we let $X = 1$ when the outcome is a success and $X = 0$ when it is a failure, then the probability mass function of X is given by

$$\begin{aligned} p(0) &= P\{X = 0\} = 1 - p \\ p(1) &= P\{X = 1\} = p \end{aligned} \tag{6.1}$$

where $p, 0 \le p \le 1$, is the probability that the trial is a success.

A random variable X is said to be a *Bernoulli random variable* (after the Swiss mathematician James Bernoulli) if its probability mass function is given by Equations (6.1) for some $p \in (0, 1)$.

Suppose now that n independent trials, each of which results in a success with probability p or in a failure with probability $1 - p$, are to be performed. If X represents the number of successes that occur in the n trials, then X is said to be a *binomial random variable* with parameters (n, p). Thus, a Bernoulli random variable is just a binomial random variable with parameters $(1, p)$.

The probability mass function of a binomial random variable having parameters (n, p) is given by

$$p(i) = \binom{n}{i} p^i (1 - p)^{n-i} \qquad i = 0, 1, \ldots, n \tag{6.2}$$

The validity of Equation (6.2) may be verified by first noting that the probability of any particular sequence of n outcomes containing i successes and $n - i$ failures is, by the assumed independence of trials, $p^i(1 - p)^{n-i}$. Equation (6.2) then follows, since there are $\binom{n}{i}$ different sequences of the n outcomes leading to i successes and $n - i$ failures. This perhaps can most easily be seen by noting that there are $\binom{n}{i}$ different choices of the i trials that result in successes. For instance, if $n = 4, i = 2$, then there are $\binom{4}{2} = 6$ ways in which the four trials can result in two successes, namely, any of the outcomes (s, s, f, f), (s, f, s, f), (s, f, f, s), (f, s, s, f), (f, s, f, s), and (f, f, s, s), where the outcome (s, s, f, f) means, for instance, that the first two trials are successes and the last two failures. Since each of these outcomes has probability $p^2(1 - p)^2$ of occurring, the desired probability of two successes in the four trials is $\binom{4}{2} p^2(1 - p)^2$.